

Organisational Climate Drivers of Employee Turnover

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1 Business understanding

1.1 What are the business objectives?

Background

ServiceFirst is a national organisation that services large equipment for corporate and private clients in Australia. They are successful business with 25 years in the industry and their company boasts 250 employees within a multi-layered structure with various departments. This includes remote service staff and office-based support teams spread across regional offices nationwide.

The Executive, People & Culture (P&C), Finance, and IT teams are located at ServiceFirst's headquarters in Sydney. In addition, there is a customer service and admin team at each regional office led by a regional manager. These managers supervise a team of remote-working service staff within their regions.

Unfortunately, during the pandemic many employees resigned which led to high recruitment costs and a loss of experience within the organisation. Therefore, the Executive team engaged the P&C team to better understand the reasons for the resignations.

The P&C team initiated a research project to establish current employee satisfaction levels and whether there is a link to resignations. They designed a questionnaire to establish how employees perceive the organisation's climate. The questionnaire included an organisational climate measure (OCM) with ten dimensions; involvement in decision-making, autonomy, relationship with direct manager, feeling part of a team, welfare, training, work-life balance and organisation's response to change, innovation and its customers (Patterson et al., 2005).

Research has shown that approximately 70% of employee dissatisfaction stems from the organisational climate, and enhancing the climate can have a direct impact on business goals (Hadi Ali et al., 2020). Hadi Ali et al. (2020) found that employees who feel supported and valued by their organisation are more likely to stay, view the organisational climate positively and see the organisation as welfare-oriented and innovative. In a study of pharmacists, Lan et al. (2019) found that organisational climate had a positive and significant influence on retention ($\beta = 0.401^*$, $P < .001$). In particular they found that those with tenure longer than 8 years had a higher intention to stay than those with a tenure of 1-8 years ($F = 4.855^{**}$, $P < .01$), and the intention to stay was greater among managers than it was among general staff ($t = -2.991^{**}$, $P < .01$). They also found a negative correlation between job stress and intention to stay ($r = -.47^{***}$, $P < .001$), and a negative correlation between workplace burnout and intention to stay ($r = -.31^{**}$, $P < .01$) (Lan et al., 2020). Overall positive perceptions of organisational climate are more likely to lead to higher individual and organisational performance.

Business objectives and success criteria

ServiceFirst's aims to reduce recruitment costs and minimise the loss of experienced employees by fostering a more positive organisational climate. Specifically, the goal of P&C team's project is to:

1. Identify the dimension(s) of the OCM where employees have scored the lowest, indicating areas of potential dissatisfaction.
2. Analyse the relationship between overall organisational climate (OC) and key factors including:
 - a. Work location, length of tenure, management responsibilities, intention to quit and employee sentiment.
 - b. The ten dimensions of OC.
3. Predict which key factors most significantly impact the overall OC, enabling the organisation to target areas for improvement, ultimately supporting increased employee retention.

Additional consultation with ServiceFirst's Executive team is required to establish specific, measurable targets for employee retention and recruitment cost reduction, along with a defined project timeline.

1.2 Assess the situation

Inventory of resources

- Data scientist pro bono
- Python language within Jupyter notebook.
- Various libraries and modules were imported. In Jupyter they were only imported only once when they were first required (refer to Table 1).

Table 1 Libraries and modules imported

Libraries and modules imported	Description
pandas as pd NumPy as np	Numpy provides a variety of mathematical functions that can perform on entire array of data without the need for loops. Some of the applications of Numpy include data cleaning (e.g., fill missing data) and providing statistical analysis. An extension of Numpy, Pandas enables importing data from a file (e.g., csv) and in turn preview, manipulate, transform and visualise that data. Numpy and Pandas are typically imported together to perform initial cleaning and exploration of data.
warnings.filterwarnings('ignore')	Suppresses warning messages that alert for potential issues in the code that might not necessarily cause it to fail but could lead to problems under certain conditions.
matplotlib.pyplot as plt seaborn as sns	These are two-dimensional plotting libraries required to conduct visualisations, including bar charts, box plots, histograms and scatterplots.
Statistical analyses	
scipy.stats as stats	Calculate means, standard error means, confidence intervals
scipy.stats import ttest_ind	Perform t-tests
statsmodels.stats.power import TTestIndPower	Calculate power of t-tests
scipy.stats import f_oneway	Perform ANOVA analysis
statsmodels.stats.multicomp import pairwise_tukeyhsd	Perform Tukey's HSD test to determine which pairing is significant in the ANOVA analysis

Libraries and modules imported	Description
statsmodels.formula.api import ols statsmodels.stats.anova import anova_lm statsmodels.stats.power import FTestAnovaPower	Calculate power of ANOVA analysis
scipy.stats import chi2_contingency	Perform chi-square analysis
Linear regression modelling	
sklearn statsmodels.api as sm sklearn.linear_model import LinearRegression	Perform linear regression modelling
sklearn.model_selection import train_test_split	Conduct data splitting
sklearn import metrics	Calculate metrics for linear regression model
statsmodels.stats.outliers_influence import variance_inflation_factor	Calculate variance inflation factor
import statsmodels.stats.api as sms from statsmodels.tools.tools import add_constant	Perform Breusch-Pagan test to statistically test for homoscedasticity

Assumptions

It is assumed that the data collected by the P&C team adhered to ethical research standards, including compliance with informed consent procedures, protection of participant confidentiality, and voluntary participation. Ethical considerations likely included transparent communication about the purpose of the data collection, secure handling of sensitive information, and ensuring that participants understood their rights, including the option to withdraw at any time without consequence. This adherence to ethical guidelines is essential to ensure the integrity of the research and protect the privacy and welfare of participants.

1.3 Data mining goals

The data mining goals relevant to the business goals are provided in Table 2.

Table 2 Data mining goal against business goal

Business goal	Data mining goal
1: Identify the dimension(s) of the OCM where employees have scored the lowest, indicating areas of potential dissatisfaction.	The OCM includes the following ten dimensions: <i>involved, autonomy, direct_manag, integrated, welfare, training, work_press, reflexive, innovative and customers</i> (the description of each of the measures is provided in Table 4). Mean scores of each of the OC dimensions will be calculated and ranked from lowest to highest scores. The dimensions with lowest scores will be highlighted as areas of potential dissatisfaction. The mean overall OC score will also be compared across demographic factors (<i>work location, length of tenure, management responsibilities, intention to quit and employee sentiment</i>) to determine if there are any differences.
2: Analyse the relationship between overall (OC) and key factors including: 2a): Work location, length of tenure, management responsibilities, intention to quit and employee sentiment. 2b): The ten dimension of OC.	For 2a) perform statistical analyses to identify any significant differences between mean overall OC score and each demographic factors. That is: • <i>Work location:</i> office vs remote • <i>Work tenure:</i> 0-5 years, 6-10 years, 11-15 years, 16-20 years and more than 20 years • <i>Management responsibility:</i> Yes vs No • <i>Intent to quit:</i> Yes vs No • <i>Sentiment:</i> Positive vs Negative For 2b) perform regression analyses to determine the strength of correlations between overall OC score and the OC dimensions and demographic variables by: • Initially performing single linear regression analyses on at least two of the OC dimensions that show high correlation with the overall OC score. • If the single linear regression modelling does not provide meaningful results, perform multiple linear regression analyses with OC dimensions and significant demographic factors to better understand the combined effects on overall OC score.
3: Determine which key factors most significantly impact the overall OC, enabling the organisation to target areas for improvement, ultimately supporting increased employee retention.	Apply predictive modelling to identify which factors most significantly impact overall OC and retention, focusing on variables with the highest predictive power.

1.4 Project plan

The data science project plan to achieve the business goals follows the Cross Industry Standard Process for Data Mining (CRISP-DM) framework and is provided in Table 3.

Table 3 Data science project plan

Plan	Description
1. Business understanding	• Define business objectives, assess the situation, and set goals for the data mining project (provided in Section 1)
2. Understand the data	• Collect and describe the data • Explore the data by performing descriptive statistics and visualisations to gain a better understanding of the key factors that impact on the overall OC measure • Examine the quality of the data • The initial exploration of the data and analyses will determine if there is any data transformation/preparation required (refer to Section 2)
3. Prepare and clean the data	• Select and clean the data to be used for analyses • Construct and combine data if required (refer to Section 3)
4. Conduct statistical modelling	• Conduct single and multiple linear regression modelling and assess their performance (refer to Section 4)
5. Evaluate the model, provide recommendations and conclusion	• Evaluate the model to ensure they meet the business goals and select the most appropriate model • Provide conclusions and recommendations based on initial data science project • Provide strategies for ServiceFirst as to how to implement the model within the organisation in order to achieve the business goals (refer to Section 5)

2 Data understanding

2.1 Initial data collection

The P&C team developed a questionnaire which included six demographic variables (*agree*, *workloc*, *workleng*, *manag_resp*, *intent_quit*, *sentiment*) and an OCM with ten dimensions (*involved*, *autonomy*, *direct_manag*, *integrated*, *welfare*, *training*, *work_press*, *reflexive*, *innovative* and *customers*). An overall OC score was provided which was the sum of the ten OC dimensions. A description of each of the items is provide in Table 4.

The P&C team surveyed 196 employees at the headquarters (office) and regional offices/teams (remote). Only 142 employees permitted their data to be used in the project.

The structured data was provided in a comma-separated values (CSV) file named *employee_survey.csv* and imported into a Jupyter Notebook for analysis in Python. The dataframe (df) was called 'survey'.

2.2 Data description

As shown below from the Jupyter notebook the original *employee_survey.csv* (*survey df*) consists of 18 columns and 142 entries.

```
RangeIndex: 142 entries, 0 to 141
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                     142 non-null   int64
1   agree                 142 non-null   object
2   workloc               142 non-null   int64
3   workleng              142 non-null   int64
4   manag_resp            142 non-null   int64
5   involved              142 non-null   float64
6   autonomy              142 non-null   float64
7   direct_manag          142 non-null   float64
8   integrated            142 non-null   float64
9   welfare               142 non-null   float64
10  training              142 non-null   float64
11  work_press            142 non-null   float64
12  reflexive             142 non-null   float64
13  innovative            142 non-null   float64
14  customers             142 non-null   float64
15  overall_score         142 non-null   float64
16  intent_quit           142 non-null   int64
17  sentiment             142 non-null   int64
dtypes: float64(11), int64(6), object(1)
memory usage: 20.1+ KB
```

As shown in the Table 4, the six demographic variables (*agree*, *workloc*, *workleng*, *manag_resp*, *intent_quit*, *sentiment*) are categorical variables and the OCM dimensions including overall score are continuous variables. All the categorical variables except *agree* have been provided as numerical data. The *agree* variable will be checked if all are 'yes' and if so the column dropped from the analyses. Therefore conversion to a numerical value was not required.

Table 4 Variables of employee_survey datafile (survey dataframe)

Variable Name	Description/Question	Options	Type of variable	Class of variable
Demographic information				
id	Unique id (anonymous)	NA	number (int64)	categorical
agree	Permission for the respondent's data to be used in the project.	yes = 1 no = 2	string (object)	categorical
workloc	Main workplace: Please choose one of the following options.	office = 1 remote = 2	number (int64)	categorical
workleng	How long have you worked at company? Please choose 1 option.	a mean score of years within each category 0-5 years = 3 6-10 years = 8 11-15 years = 13 16-20 years = 18 More than 20 years= 23	number (int64)	categorical (ordinal)
manag_resp	I have line management responsibility for employees	Yes = 1 No = 2	number (int64)	categorical
Organisational climate measure (10 of variables of the Patterson et al. 2005 measure)				
involved	Do you feel involved in organisational decision-making?	Standardised scores ranging from -5 (very dissatisfied) to +5 (very satisfied)	number (float64)	continuous
autonomy	Do you feel that you are trusted to do your job independently?			
direct_manag	Do you have a positive working relationship with your direct manager?			
integrated	Do you feel part of your team?			
welfare	Do you feel that the organisation cares about you?			
training	Do you get enough training to do your job effectively?			
work_press	Is your work-life balance manageable?			
reflexive	Does the organisation respond well to externally driven change?			
innovative	Do the organisation welcome ideas for innovation and implement these?			
customers	Does the organisation respond well to customer needs?			

Variable Name	Description/Question	Options	Type of variable	Class of variable
overall_score	Overall satisfaction score (sum of the 10 standardised scores for each response)	Overall score ranging from -50 to +50 The higher the number, the more satisfied the employee is overall	number (float64)	continuous and/or categorical (ordinal)
Intention to leave and sentiment measures				
intent_quit	Do you intend leaving the organisation in the next six months?	Yes = 1 No = 2 Not sure = 3	number (int64)	categorical
sentiment	Sentiment codes for textual responses.	positive = 1 negative = 2 no response = 3	number (int64)	categorical

2.3 Data quality report

To assess the quality of the data and rectify issues if required, the following steps were performed:

- Determine if there are any missing data,
- Determine if there are any duplicate data,
- Determine incorrect values by examining unique values of the categorical data, and confirm that overall_score equals the sum of the ten OC dimensions,
- Determine if there are any outliers.

There was no missing data, duplicate data or incorrect values found. In addition, the overall_score provided in the original dataset matched the calculate_score. Refer to Jupyter Notebook. Outliers and anomalies in the continuous variables were checked during data exploration by performing descriptive statistics and examining distributions through visualisations. Refer to Section 2.4.

2.4 Exploration report

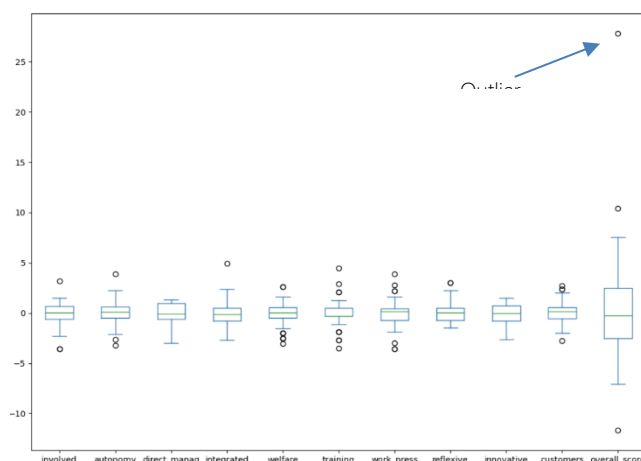
The data was explored as follows:

- For continuous data variables, the presence of outliers was investigated and normality of distributions was confirmed by:
 - Creating visualisations of the continuous variables via box plots and histograms
 - Calculating the skewness and kurtosis
- Calculating descriptive statistics of continuous and categorical data
- To address business goal 1, reviewing differences of the means of the various OC dimensions and rank from lowest to highest; reviewing means of the overall OC score and each of the demographic variables.
- To address business goal 2 and 3, explore the correlations between the variables

Outlier

Only one outlier was identified (refer to box plots in Figure 1) and was subsequently removed to prevent it from skewing the data analysis. Without removing this outlier, the mean values for all OC dimensions, including the overall score, were effectively zero to three decimal places, making it challenging to derive meaningful insights from data analysis.

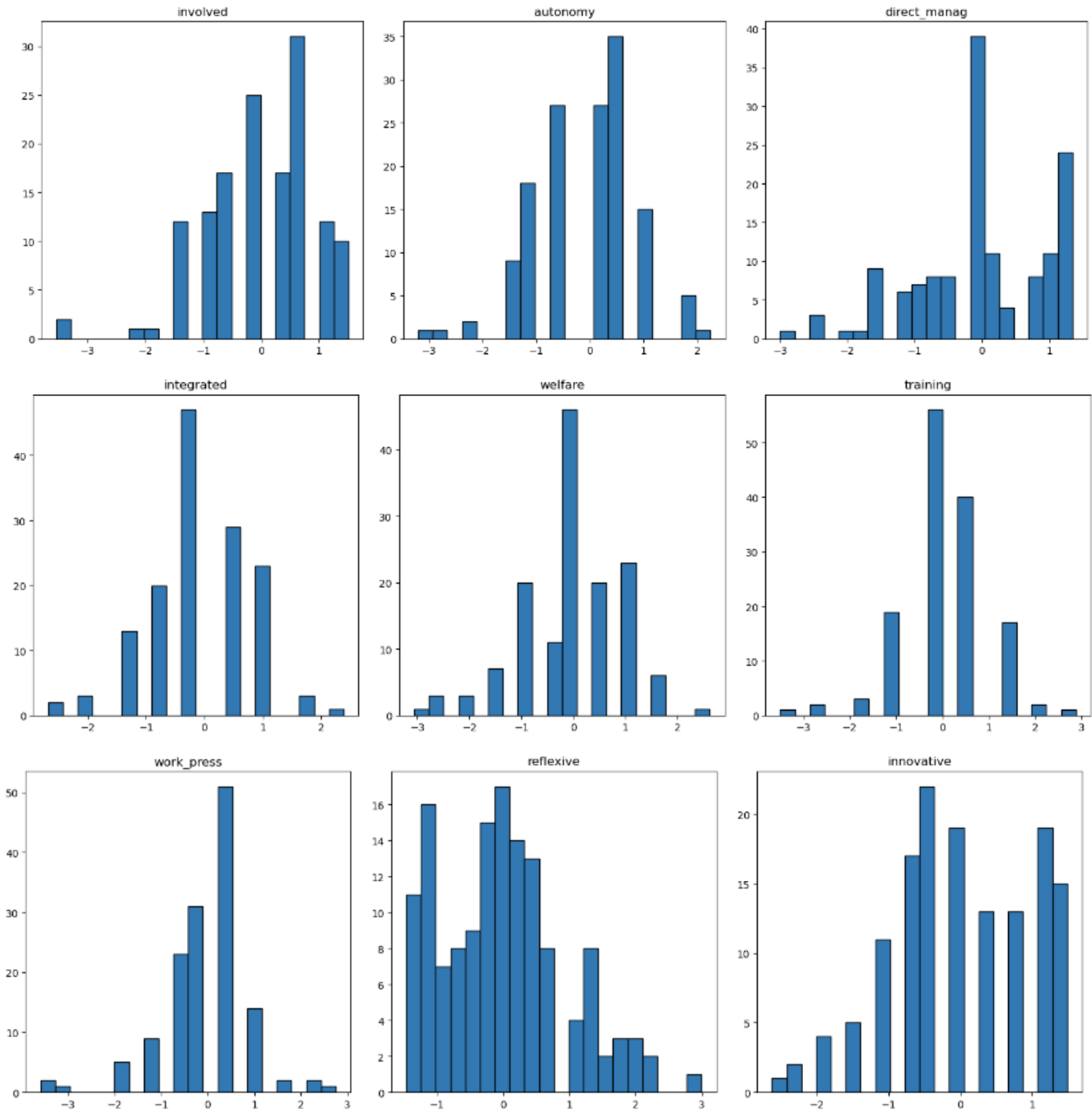
Figure 1 Box plots for variables overall_score and OC dimensions

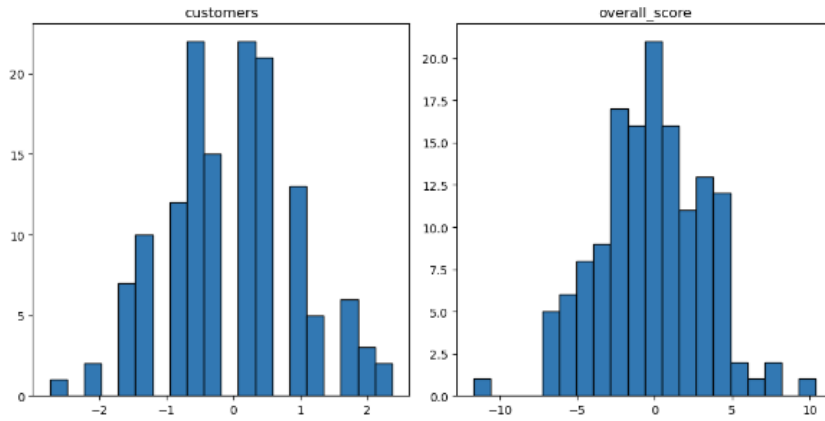


Normal distribution

The histogram, skewness, and kurtosis analyses indicate that the overall OC score and the OC dimensions are approximately normally distributed (see Figure 2). Skewness values fell between -1 and +1, suggesting low skewness overall. The variables *involved*, *autonomy*, *direct_manag*, *integrated*, *welfare*, *training*, *work_press*, and *innovative* have skewness values between -1 and 0, indicating slight left skewness. The variable *reflexive* shows slight right skewness, while *customers* and *overall_score* exhibit virtually no skewness. Kurtosis values are also low and close to 0 for most variables, except for *training* and *work_press*, which have kurtosis values of 1.69 and 2.71, respectively, indicating slightly more peaked distributions. The assumption of normality has been verified for conducting statistical analyses on the continuous variables particularly regression analyses.

Figure 2 Histograms for variables overall_score and OC dimensions





Descriptive statistics Continuous variables

Table 5 provides the mean, standard deviation, standard error of mean, 95% confidence interval (CI) and interquartile range (IQR) of each of the variables in the OCM. The OC dimensions have been ranked from lowest to highest mean score.

Table 5 Descriptive statistics of the variables in the OCM

	mean	Standard deviation	Standard error of mean	Lower CI	Upper CI	Quartile 1 (Q1)	Quartile 2 (Q2)	IQR
overall_score	-0.198	3.443	0.290	-0.771	0.375	-2.520	2.440	4.960
integrated	-0.035	0.913	0.077	-0.187	0.117	-0.781	0.486	1.267
training	-0.032	0.929	0.078	-0.186	0.123	-0.309	0.489	0.797
work_press	-0.028	0.947	0.080	-0.185	0.130	-0.706	0.446	1.152
autonomy	-0.028	0.948	0.080	-0.185	0.130	-0.475	0.613	1.089
involved	-0.023	0.966	0.081	-0.184	0.138	-0.596	0.668	1.264
reflexive	-0.021	0.971	0.082	-0.183	0.140	-0.723	0.517	1.240
customers	-0.019	0.976	0.082	-0.182	0.143	-0.550	0.545	1.095
welfare	-0.019	0.979	0.082	-0.182	0.144	-0.477	0.557	1.033
direct_manag	-0.010	0.997	0.084	-0.176	0.156	-0.683	0.764	1.447
innovative	0.016	0.985	0.083	-0.148	0.180	-0.756	0.745	1.501

The results indicate that the overall OC score is slightly negative, with a 95% CI ranging from -0.771 and 0.375 and an IQR of 4.960, suggesting a substantial variation of the climate score. Most OC dimensions have mean scores near zero, ranging from -0.035 (integrated) to 0.016 (innovative). The 95% CIs cross zero, suggesting that true mean scores may be slightly negative or positive. This issue may partly result from the P&C team's Likert scale ranging from -5 (very dissatisfied) to +5 (very satisfied). A likert scale from 1 (very dissatisfied) to 5 (very satisfied) may have provided more meaningful averages. A larger sample size could have further narrowed CI, clarifying true climate scores.

Despite this issue, the dimensions *integrated* and *training* scored the lowest suggesting that employees feel less part of the team and lack sufficient training to perform their roles effectively.

Categorical variables

Bar charts showing the proportion of values for each demographic categorical variable are presented in Figure 3 to 7.

Figure 3 Work location

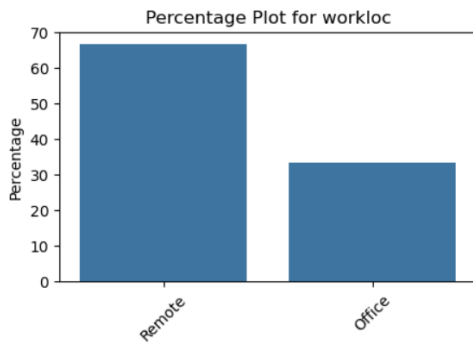


Figure 6 Intent to quit

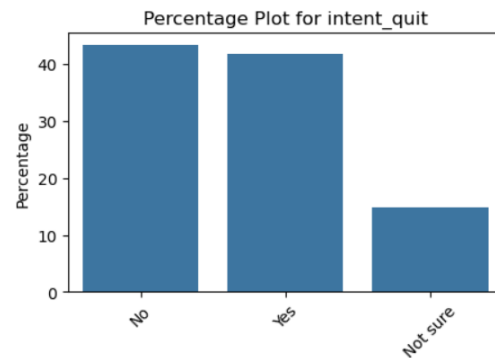


Figure 4 Work length

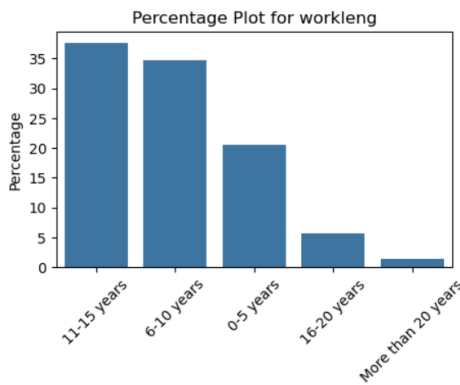


Figure 7 Sentiment

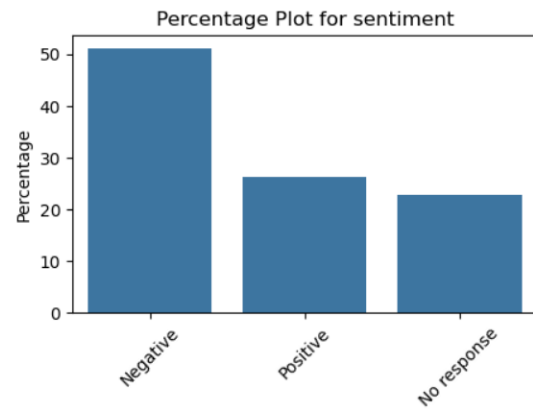
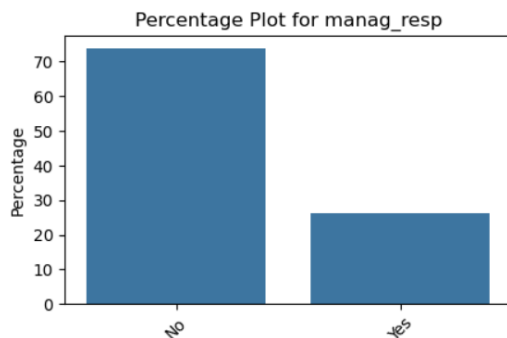


Figure 5 Management responsibility



The majority of employees that were surveyed were from remote locations (67%) and did not have line management responsibility of employees (74%). Most employees have been with the organisation either 6-10 years (35%) or 11-15 years (38%). The proportion of employees intending to quit in the next 6 months (43%) was very similar to the proportion not intending to quit (42%). In contrast, more employees had negative responses to textual responses (51%) compared to positive (26%) or neutral (26%).

The proportions of intent_quit values (yes, no, not sure) were calculated for each of the demographic variables (workloc, workleng, manag_resp and sentiment (refer to Table 6). The highest values are highlighted in yellow. More employees intend to quit in the next six months if they were located in the office, if they worked over 11 years, if they had management responsibility or if they scored negatively in textual responses.

Table 6 Proportions of intent_quit for each demographic variable

Crosstab between intent to quit and workloc:

workloc	Office	Remote
intent_quit		
No	40.4	44.7
Not sure	10.6	17.0
Yes	48.9	38.3

Crosstab between intent to quit and workleng:

workleng	0-5 years	11-15 years	16-20 years	6-10 years	\
intent_quit					
No	65.5	34.0	25.0	44.9	
Not sure	17.2	15.1	12.5	14.3	
Yes	17.2	50.9	62.5	40.8	

workleng More than 20 years

intent_quit	
No	0.0
Not sure	0.0
Yes	100.0

Crosstab between intent to quit and manag_resp:

manag_resp	No	Yes
intent_quit		
No	44.2	40.5
Not sure	16.3	10.8
Yes	39.4	48.6

Crosstab between intent to quit and sentiment:

sentiment	Negative	No response	Positive
intent_quit			
No	13.9	43.8	100.0
Not sure	5.6	53.1	0.0
Yes	80.6	3.1	0.0

Continuous vs categorical variables

The overall OC climate means were compared for each of the categories of the demographic variables and provided in Table 7. Lowest values are highlighted in yellow.

Table 7 Overall OC climate means for each demographic variable

Average Overall Score for workloc:

workloc	mean_overall_score	std_deviation	std_error
0 Office	-0.413404	3.425337	0.499637
1 Remote	-0.090000	3.464423	0.357328

Average Overall Score for workleng:

workleng	mean_overall_score	std_deviation	std_error
0 0-5 years	-0.170690	3.620924	0.672389
1 11-15 years	0.766226	3.557798	0.488701
2 16-20 years	-0.521250	3.672401	1.298390
3 6-10 years	-1.295510	2.947753	0.421108
4 More than 20 years	2.050000	1.767767	1.250000

Average Overall Score for manag_resp:

manag_resp	mean_overall_score	std_deviation	std_error
0 No	-0.031346	3.472202	0.340477
1 Yes	-0.665676	3.360024	0.552385

Average Overall Score for intent_quit:

intent_quit	mean_overall_score	std_deviation	std_error
0 No	-0.336721	3.544547	0.453833
1 Not sure	-0.308571	3.776166	0.824027
2 Yes	-0.014746	3.260977	0.424543

Average Overall Score for sentiment:

sentiment	mean_overall_score	std_deviation	std_error
0 Negative	-0.034583	3.185865	0.375458
1 No response	-0.253437	3.539960	0.625782
2 Positive	-0.467297	3.892276	0.639886

Those working in the office have lower overall OC scores than those working remotely (where less employees intent to quit,). Employees with longer tenure (more than 20 years) have a notably positive mean overall score ($M=2.05$, $SD=1.76$), while employees in the 6-10 year range have the most negative score ($M=-1.30$, $SD=2.95$). This may suggest that employees newer to the company (0-5 years) and those with the longest tenure tend to view the organisational climate more positively, while mid-tenure employees, especially those with 6-10 years, have more negative perceptions. Employees with management responsibilities have a slightly more negative mean score ($M=-0.666$, $SD=3.36$) compared to those without ($M=-0.031$, $SD=3.47$). This could indicate that managers experience more challenges or have a more critical perception of the organisational climate. Interestingly, those intending to quit do not have the most negative perception, which might suggest other factors driving their intent to leave beyond organisational climate. It is counterintuitive that employees with positive sentiment scores have the most negative overall OC score. This discrepancy may be due to differences in how sentiment is coded from textual responses, as opposed to the OC score, which is based on the OCM adapted from a reliable and validated measure.

Statistical analyses will be conducted to assess whether there are significant differences in mean overall OC scores across the categories of each demographic variable, and to determine if the proportions for intent to quit differ significantly across these demographic categories (refer to Section 4).

Correlations between variables

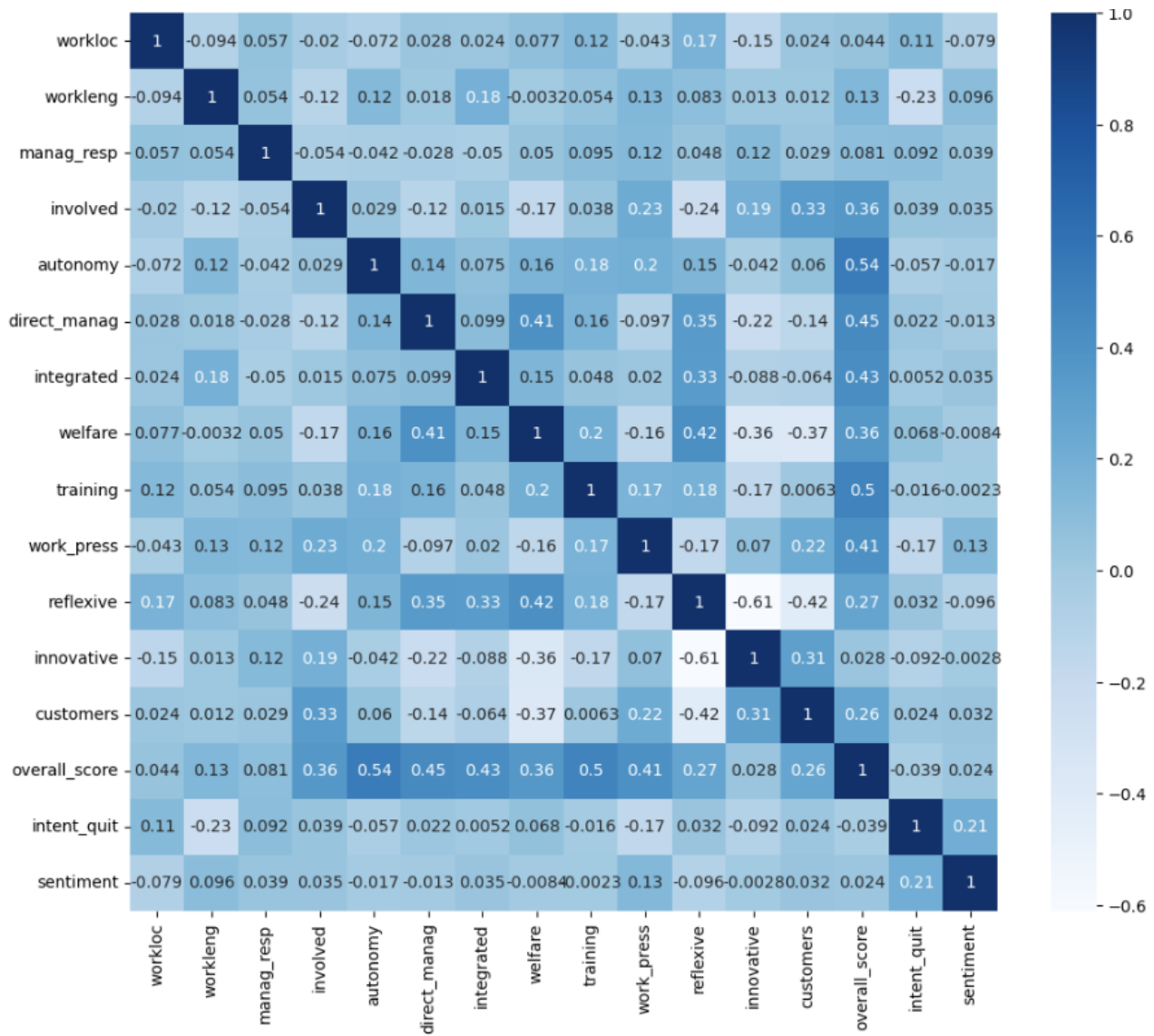
Refer to the heatmap in Figure 8, which displays the correlation coefficients (Pearson's r), quantifying the strength and direction of the relationship between pairs of variables in the regression analysis. The following was found:

- *overall_score* of OC measure has a very weak to almost no correlation ($-0.044 \leq r \leq 0.081$) with the variables *workloc*, *manag_resp*, *intent_quit*, *sentiment* and one of the OC dimension *innovative*.
- *overall_score* has a very weak positive correlation ($r=0.13$) with the variable *workleng*
- *overall_score* has a weak to moderate positive correlation ($0.26 \leq r \leq 0.36$) with the OC dimensions *involved*, *welfare*, *reflexive* and *customers*.
- *overall_score* has a substantial to strong positive correlation ($0.41 \leq r \leq 0.54$) with the OC dimensions *autonomy*, *direct_manag*, *integrated*, *training* and *work_press*.

There are also positive or negative moderate to strong correlations between the OC dimensions.

- For *reflexive*, there is a moderate positive correlation with *direct_manag* and *integrated* ($r=0.35$ and $r=0.33$ respectively); medium positive correlation with *welfare* ($r=0.42$); medium negative correlation with *customer* ($r=-0.42$) and strong negative relationship with *innovative* ($r=-0.61$)
- For *welfare*, there is a medium positive correlation with *direct_manag* ($r=0.41$)

Figure 8 Heatmap of correlations between variables



These results will inform the linear regression modelling to address the business goals (refer to Section 4).

3 Data preparation

3.1 Select data

The dataframe utilised to conduct the hypothesis testing/modelling included all the columns of the original dataset except *id* and *agree* as they both do not provide any meaningful information about employee's characteristics or the variables of interest. In addition the *calculated_score* (sum of the OC dimensions), that was calculated to compare with *overall_score* and check for anomalies during the data exploration was also dropped.

When exploring the distribution of continuous variables, only the variables of the OCM were included. For the single linear regression modelling only the variables that have the highest correlation with the overall OC score were examined. In summary the dataframes within Jupyter were labelled as follows:

- survey = full data set provided ('employee_survey.csv') (142 rows, 18 columns)
- survey_c = includes only continuous variables (*involved*, *autonomy*, *direct_manag*, *integrated*, *welfare*, *training*, *work_press*, *reflexive*, *innovative*, *customers*, *overall_score*) used during the exploration of the continuous variables.
- survey_a = is the clean data set and includes all variables except the unnecessary variables *id*, *agree* and *calculated score* (141 rows, 16 columns).
- survey_s = includes only the variables for the single linear regression modelling (141 row, 3 columns).

3.2 Clean data

Please refer to Section 2.3 and 2.4.

3.3 Construct data

To check for anomalies in *overall_score*, a *calculated_score* column was created as follows:

- $Calculated_score = involved + autonomy + direct_manag + integrated + welfare + training + work_press + reflexive + innovative + customers$

This was compared with *overall_score* and no anomalies were found.

3.4 Integrate and formatting

No integration and re-formatting was required except when the values of the categorical variables were relabelled with the original names for ease of readability of tables and visualisations. However, they were reverted back to numerical values prior to hypothesis testing and modelling.

4 Hypothesis testing and modelling

The following hypothesis testing and modelling were conducted:

4.1 Mean overall_score and demographic categorical variables:

As identified during data exploration, differences were observed in the mean *overall_score* between the categories of the demographic variables. Therefore, the following hypothesis was proposed:

a. Hypothesis

H_0 : There is no difference in the mean *overall_score* between the categorical values of each demographic variable (i.e., group1 (value= 1) = group 2 (value= 2)).

H_a : There is a significant difference in the mean *overall_score* between the categorical values of each demographic variable (i.e., group1 (value= 1) $\neq$ group 2 (value= 2)).

b. Statistical test utilised

For the variables *workloc* (1=office; 2=remote), *manag_resp* (1=yes, 2=no), *intent_quit* (1=yes, 2=no), *sentiment* (1=positive, 2=negative), the independent t-test (two-tailed) was utilised as the means of two independent groups will be compared and the standard deviation of the population mean was unknown. A significance level (α) of 0.05 was utilised which is a common level for research projects.

For the variable *workleng* which has five categories, the one-way ANOVA (Analysis of Variance) test was utilised to determine whether there are statistically significant differences between the means of three or more independent groups. A significance level (α) of 0.05 was utilised. If any significant difference was found, the Tukey's HSD test was utilised to determine which pairing was significant.

The assumptions for both the independent t-test and one-way ANOVA are that the observations in each group are independent, the data within each group are approximately normally distributed, and the variances across groups are approximately equal. Data exploration confirmed normality of distributions and approximate equality of variances (see Figure 2 and Table 5, Section 2.4).

The effect size and power of the statistical tests above were also examined.

c. Results

The output for the independent t-test and power analysis is provided in Table 8.

Table 8 Independent t-test and power analysis output

	Independent t-test: Statistic	p-value	Cohen's d (effect size)	Power of the test
workloc	-0.526491	0.599800	-0.093699	0.081616
manag_resp	-0.977566	0.331899	-0.184211	0.159398
intent_quit	0.518103	0.605359	0.094473	0.080675
sentiment	-0.583247	0.561864	-0.125797	0.094528

For each of the variables (*workloc*, *manag_resp*, *intent_quit*, and *sentiment*), the p-values are all greater than 0.05. Therefore, we cannot reject the null hypothesis for all the tests. Therefore, there is no statistically significant difference in the mean overall OC score between the two categorical groups for any of these variables. The effect size (Cohen's d) for all the variables are close to zero, indicating that means of the two groups differ by less than 0.2 standard deviations, which is a negligible difference.

However, the tests have very low power (8% to 16%), indicating a high chance of Type II errors and insufficient sensitivity to detect small effects. Increasing the sample size could improve power, making it easier to detect smaller effects if they exist.

For *workleng*, there was a significant difference in the mean *overall_score* between at least two of the categories ($F = 2.633$, $P < 0.05$), with a 74% chance of correctly rejecting the null hypothesis. Specifically, there is a significant difference between those who have stayed for 13 years compared to those who have stayed for 8 years ($P < 0.05$), with a mean difference of 2.0617 and a confidence interval of [0.218, 3.9055].

4.2 Intent_quit proportions and other demographic categorical variables

As identified during data exploration, differences were observed in the *intent_quit* proportions between the categories of the demographic variables. Therefore, the following hypothesis was proposed:

a. Hypothesis

H_0 : There is no difference in the *intent_quit* proportions between the categorical values of each demographic variable (i.e., group1 (value= 1) = group 2 (value= 2)).

H_a : There is a significant difference in the *intent_quit* proportions between the categorical values of each demographic variable (i.e., group1 (value= 1) $\neq$ group 2 (value= 2)).

b. Statistical test utilised

A chi-squared test was utilised to determine if there is a statistically significant relationship between the categorical variables. It is a non-parametric test, meaning it doesn't assume a normal distribution of the data. A significance level (α) of 0.05 was utilised.

The effect size and power of the chi-squared test above were also examined.

c. Results

The output for the chi-squared test is provided in Table 9.

Table 9 Chi-squared test output

	Chi-squared	P-value	Degrees of Freedom
<i>workloc</i>	1.836	0.399	2.0
<i>workleng</i>	14.030	0.081	8.0
<i>manag_resp</i>	1.202	0.548	2.0
<i>sentiment</i>	135.756	0.000	4.0

The results indicate that *workloc* and *manag_resp* have p-values greater than 0.05, suggesting no statistically significant difference in the proportions of *intent_quit* across their categories. However, the power of these tests were very low (<9%), indicating a high likelihood of Type II errors and insufficient sensitivity to detect small effects (i.e., a high probability of not rejecting the null hypothesis even if a true effect exists).

Workleng has a p-value close to 0.05, indicating a borderline association that is not quite significant at the 5% level but may warrant further investigation, especially as the test power is only 37%.

On the other hand, *sentiment* (Chi-squared = 135.756, $p < 0.001$, power = 99.9%) shows a highly significant difference in the proportions of *intent_quit*, indicating a strong association between sentiment

and intent to quit. Specifically, those with negative sentiment are more likely to express an intent to quit within the next six months.

4.3 Single Linear regression modelling

As identified during data exploration, positive correlations of various strengths were found between *overall_score* (dependent variable) and OC dimensions and one of the categorical variable *workleng* (independent variables).

The first step in the modelling is to conduct separate single linear regression models between the *overall_score* and the OC dimensions *autonomy* and *training* (as they had the highest correlations) as follows:

a. Hypothesis 1

H_0 : There is no relationship between *overall_score* and *autonomy*.

H_a : There is a positive correlation between *overall_score* and *autonomy* where the higher the autonomy (feel trusted to do their job independently) the higher the overall score of the OCM. Thus *autonomy* can accurately predict *overall_score*.

Hypothesis 2

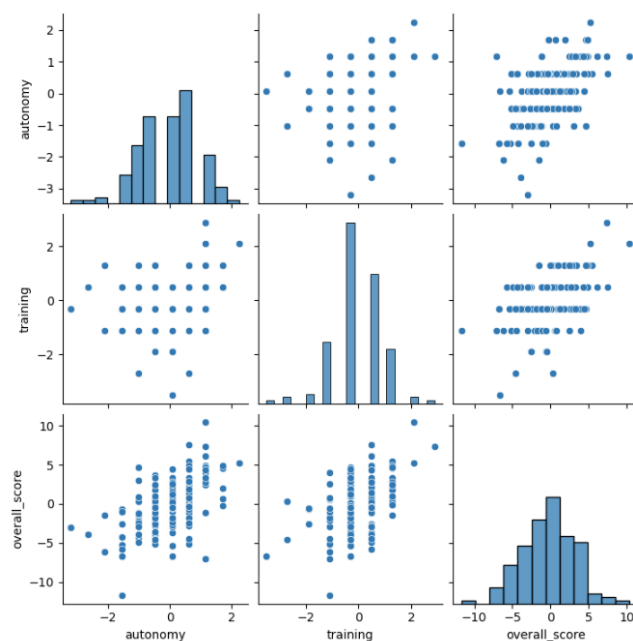
H_0 : There is no relationship between *overall_score* and *training*

H_a : There is a correlation between *overall_score* and *training* where the higher the training (feel they get enough training) the higher the overall score of the OCM. Thus *training* can accurately predict *overall_score*.

b. Modelling and test design

Simple linear regression modelling was conducted to understand and predict the relationship between the two continuous variables. However, to ensure valid results, the following assumptions needed to be met: linearity, homoscedasticity and normality of residuals. The pairplot shown in Figure 9, confirms that *autonomy*, *training* and *overall_score* have a normal distribution and there is a linear relationship between *autonomy* and *overall_score* as well as *training* and *overall_score*.

Figure 9 Pairplot of *autonomy*, *training* and *overall_score*



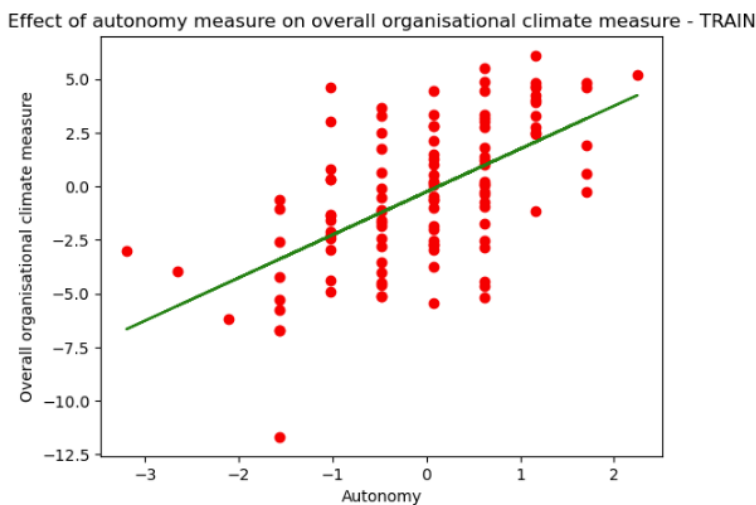
For the linear regression modelling the data was split 80:20. This split leaves 80% of the data (around 113 observations) for training and 20% (around 28 observations) for testing. With a small dataset the 80:20 split can ensure that the model has enough training data to capture patterns.

Autonomy

c. Accuracy of the model

Figure 10, provides the scatterplot for the effect of autonomy on overall_score for the training data, which confirms the linear relationship, that is the higher the autonomy the higher the overall_score. In addition, the data points are evenly grouped around the best-fit line, indicating homoscedasticity.

Figure 10 Effect of autonomy measure on overall_score for training data



With a data split of 80:20, the following metrics were obtained for autonomy:

- Coefficient of determination (R^2)=0.343: The goodness of fit is approximately 34.3%, indicating that only 34% of the variability of overall_score (Y) is explained by the autonomy measure (X). This suggests that the model may not be capturing the data's patterns well. As the relationship between the variables was previously found to be strong then there are other important variables that may be missing from the model.
- Intersection (β_0)=-0.254: The value of overall_score (Y) when autonomy measure (X) is equal to zero.
- Coefficients/slopes (β_1)=2.00: The value describes the direction and strength of the relationship between the two variables. It indicates that for each autonomy measure (X) unit, the overall_score (Y) will increase by 2.00 units.
- Mean absolute error (MAE)=0.81: MAE indicates the average absolute difference between predicted values and observed values. In this case, on average, predictions are off by about 0.81 units.
- Mean Square Error (MSE)=2.71: MSE is the average squared difference between the predicted and actual values, suggests the model's predictions are on average 2.71 units squared away from the actual values.

- Root Mean Square Error (RMSE)=1.03: The RMSE is the square root of the MSE, giving an error metric in the same units as the data. The model's predictions deviate from the actual values by about 1.03 units.

Overall, the model explains only a small portion (34.3%) of the variance in the data. The predictions on average deviate from the actual values by approximately 0.81 (MAE) and 1.03 (RMSE). Although MAE and RMSE values are low, these errors are significant when we take into account that the small values for the overall_score (Y). This suggests that the model may not be a good fit and might benefit from more variables for improved accuracy.

d. Evaluation of the model

The model was evaluated using the test data and the following outputs were found (Figure 11 and Table 10).

Figure 11 Effect of autonomy measure on overall_score for training data

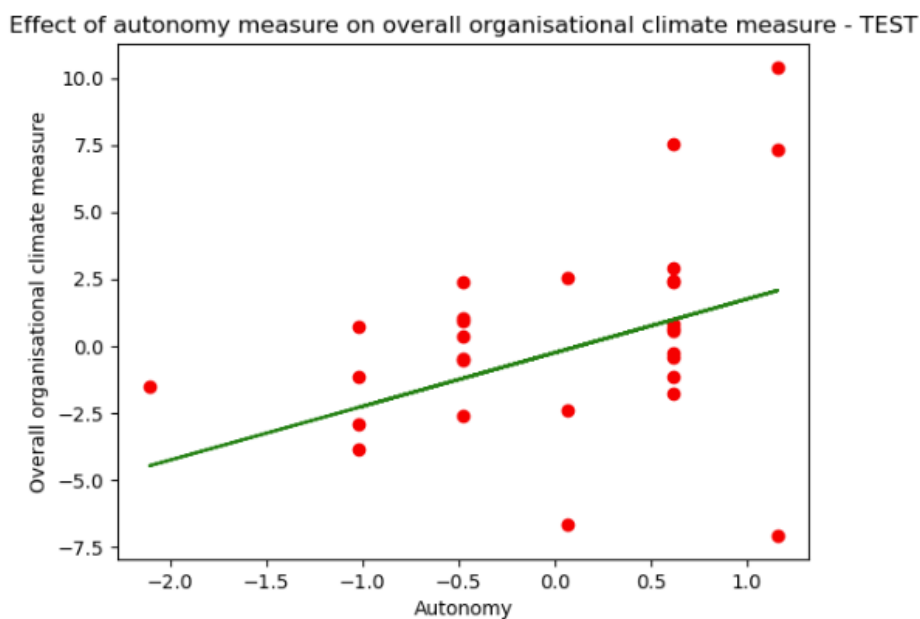


Table 10 Single regression metrics for autonomy and overall_score – test data

```
R-squared value: 0.09723633092337958
Intercept value: [-0.25403053]
Coefficient value: [[2.00286063]]
Mean Absolute Error: 0.6841390617486586
Mean Square Error: 3.489068178805746
Root Mean Square Error: 0.8278755375223099
```

Unfortunately, the regression metrics of the test data do not match the regression metrics of the training data. This indicates that the single linear regression modelling for overall_score with autonomy was not found to be accurate, that is autonomy does not accurately predict overall_score on its own and there may be other variables that predict overall_score.

Training

c. Accuracy and evaluation of the model

Similar results were found when performing single linear regression modelling for *overall_score* and training. The linear regression outputs for the training and test data are shown in Table 11 and 12.

Table 11 Single regression metrics for training and *overall_score* – train data

```
R-squared value: 0.1865571893200869
Intercept value: [-0.31050286]
Coefficient value: [[1.7245604]]
Mean Absolute Error: 0.5259617239680245
Mean Square Error: 3.021701712031409
Root Mean Square Error: 0.6883533885674251
```

Table 12 Single regression metrics for training and *overall_score* – test data

```
R-squared value: 0.40279771658090147
Intercept value: [-0.31050286]
Coefficient value: [[1.7245604]]
Mean Absolute Error: 0.7267325802173188
Mean Square Error: 2.8378089185837623
Root Mean Square Error: 0.9476659844785157
```

The R^2 values in both the training (0.187) and test (0.403) datasets are relatively low and don't match, suggesting that *training* explains a limited portion of the variance in *overall_score*. That is, *training* alone may not be sufficient to predict *overall_score* accurately.

Although both *autonomy* and *training* have a positive effect on *overall_score* (as indicated by the positive coefficient), their predictive power is limited. As additional predictors may be needed to improve the model's accuracy in explaining variations in *overall_score*, multiple linear regression modelling will be conducted.

4.4 Multiple Linear regression modelling

As identified during data exploration, there were weak or no correlations between *overall_score* and the categorical variables *workloc*, *manag_resp*, *intent_quit*, *sentiment* and one of the OC measure *innovative*. Additionally, *reflexive* was found to correlate with the many other OC measures. Based on these observations, the following hypothesis is proposed:

a. Hypothesis

H_0 : There is no relationship between *overall_score* and any of the predictor variables

H_a : There is a significant correlation between *overall_score* and at least one of the independent variables (predictors) *workleng*, *involved*, *autonomy*, *direct_manag*, *integrated*, *welfare*, *training*, *work_press*, *reflexive*, *innovative* and *customers*. This implies that one or more predictor variables contribute to predicting the *overall_score*.

b. Modelling and test design

Multiple linear regression modelling will be conducted to evaluate the combined effect of the independent variables *workleng*, *involved*, *autonomy*, *direct_manag*, *integrated*, *welfare*, *training*, *work_press*, *reflexive*, *innovative* and *customers* on the single dependent variable (*overall_score*). Multiple linear regression will help determine which variables significantly predict *overall_score* while controlling for the effects of other variables. The assumptions of multiple linear regression are as follows:

- 1. Linearity:** The relationship between the dependent variable and independent variables should be linear. This means that the change in the dependent variable should be proportional to the change in the independent variables.
- 2. Independence:** The observations should be independent of each other. This means that there should be no correlation or association between the independent variables.
- 3. No multicollinearity:** There should be no multicollinearity between the independent variables. This means that the independent variables should not be too highly correlated with each other.
- 4. Homoscedasticity:** The residuals should have a constant variance across all levels of the independent variables. This means that the spread of the residuals should be the same for all values of the independent variables.
- 5. Normality: The residuals** (the differences between the predicted and actual values of the dependent variable) should be normally distributed.

For the linear regression modelling the data was split 80:20. This split leaves 80% of the data (around 113 observations) for training and 20% (around 28 observations) for testing. With a small dataset the 80:20 split can ensure that the model has enough training data to capture patterns.

c. Checking assumptions

As identified in the heatmaps (refer to Figure 8, Section 2.4) and the pairplots reveal there are virtually weak to no correlation between overall score and the variables *workloc*, *manag_resp*, *intent_quit*, *sentiment* and one of the organisational climate measure innovative. Therefore, these would be potentially removed from the MLR. In addition, it was found that *intent_quit* and *sentiment* correlate (refer to Section 4.2) which may affect the modelling. As *reflexive* was correlated with a number of the independent variables this would also potentially be removed from the MLR. However, all these variables were not dropped until multicollinearity was confirmed with variance inflation factor (VIF) test for each dependent variable. The assumption of multicollinearity, homoscedasticity and normality of residuals are confirmed and discussed in the next two sections.

d. Accuracy of the model

Two multiple linear regression modelling were performed. The first model included the variables *workleng*, *involved*, *autonomy*, *direct_manag*, *integrated*, *welfare*, *training*, *work_press*, *reflexive*, *innovative*, *customers* and *intent_quit* as the VIF were below 5 (refer to Table 13).

Table 13 VIF for first model

	feature	VIF
0	<i>workleng</i>	3.050781
1	<i>involved</i>	1.213325
2	<i>autonomy</i>	1.148153
3	<i>direct_manag</i>	1.283636
4	<i>integrated</i>	1.194325
5	<i>welfare</i>	1.539434
6	<i>training</i>	1.141145
7	<i>work_press</i>	1.246437
8	<i>reflexive</i>	2.292836
9	<i>innovative</i>	1.728161
10	<i>customers</i>	1.440969
11	<i>intent_quit</i>	3.047686

However, the modelling results indicated overfitting, with $R^2 = 1.00$ on the training data. Therefore, in the second model, variables with a VIF > 1.5 were removed one at a time to address multicollinearity (refer to

Table 14). The decision to use a stricter VIF threshold (reducing the upper limit from 5 to 1.5) may be due to the small and near-zero values of the variables in the data, which can increase sensitivity to multicollinearity.

Table 14 VIF for first model

	feature	VIF
0	workleng	1.011702
1	involved	1.173931
2	autonomy	1.114439
3	direct_manag	1.233384
4	integrated	1.032665
5	welfare	1.474493
6	training	1.115633
7	work_press	1.175559
8	customers	1.307257

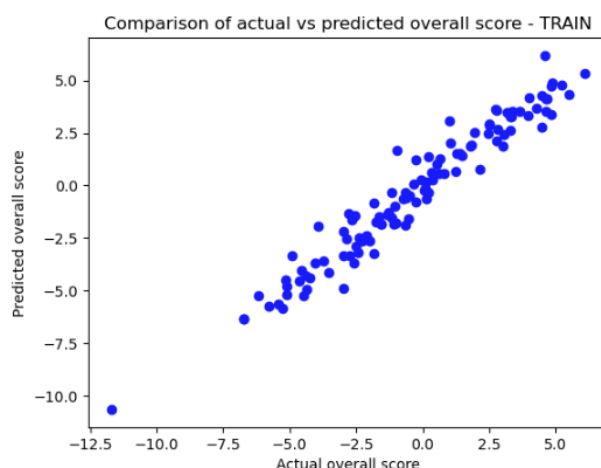
The assumption of no multicollinearity has been confirmed with the VIF testing (Table 14).

With a data split of 80:20, the following results were obtained for the train data:

- $R^2=0.945$ and adjusted $R^2=0.941$: The goodness of fit is approximately 94.1%, which indicates a very strong positive correlation between overall_score (Y) and the independent variables (Xs).
- $\beta_0=-0.0465$: The value of overall_score (Y) when all the independent variables (Xs) are equal to zero.
- $\beta_n: 0.007, 1.023, 1.095, 1.091, 1.234, 0.895, 1.013, 0.954, 0.8785$. These values describe the direction and strength of the relationship between the overall_score (Y) and each of the independent variables (Xs)
- MAE and MSE: The respective values are 0.584 and 0.613.
- RMSE: The value is 0.783.

The scatterplot of the train data is provided in Figure 12.

Figure 12 Comparison of actual vs predicted overall score for training data



e. Evaluate the model

The model was evaluated with the test data and it was found to be accurate with regression metrics being very similar to the training data (refer to Table 15).

Table 15 Multiple linear regression metrics for overall_score – test data

R-squared value (test): 0.9405443952805324
Adjusted R-squared value (test): 0.9123812140976266

Intercept value: (test) -0.04650248472553414

Coefficient value: (test) [0.00749067 1.02332431 1.09492316 1.09073571 1.2348358 0.89475984
1.01376983 0.95446143 0.87859316]

Mean Absolute Error (test): 0.7243395623148601

Mean Square Error (test): 0.8017475465094249

Root Mean Square Error (test): 0.8954035662813863

In addition, it was found that there were very small difference between the actual and predicted overall_score values. The model also met the assumption of homogeneity (where the spread of the residuals was constant, refer to Figure 13 and Breush-Pagan test had a p-value >5), and the assumption that the residuals are normally distributed (refer to Figure 14).

Figure 13 Homogeneity of residuals

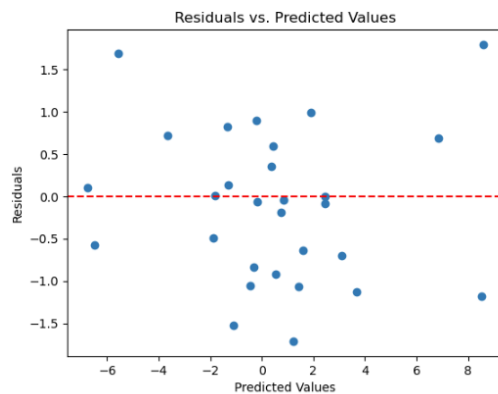
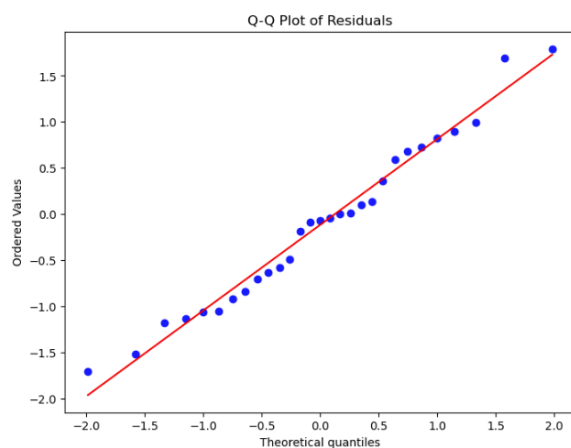


Figure 14 Normality of residuals



f. Making predictions

The final MLR model is as follows:

- $\text{overall_score} = -0.0465 + (0.0075 \times \text{workleng}) + (1.0233 \times \text{involved}) + (1.0949 \times \text{autonomy}) + (1.0907 \times \text{direct_manag}) + (1.2348 \times \text{integrated}) + (0.8948 \times \text{welfare}) + (1.0138 \times \text{training}) + (0.9545 \times \text{work_press}) + (0.8786 \times \text{customers})$

and has been rewritten below to place the independent variables in descending order of their influence on overall_score.

- $\text{overall_score} = -0.0465 + (1.2348 \times \text{integrated}) + (1.0949 \times \text{autonomy}) + (1.0907 \times \text{direct_manag}) + (1.0233 \times \text{involved}) + (1.0138 \times \text{training}) + (0.9545 \times \text{work_press}) + (0.8948 \times \text{welfare}) + (0.8786 \times \text{customers}) + (0.0075 \times \text{workleng})$

According to the model, integrated has the most influence on the overall score, followed by autonomy and direct_manag.

To illustrate the influence of the independent variables on the overall OC score, the following scenario were provided:

- Two groups of length of work (lowest and highest values):
 - group1, workleng=3 (length at company 0 to 5 years)
 - group2, workleng=23 (more than 20 years)
- For each group the following options of the independent variables were investigated:
 - Option0: all variables = 0 (neutral, neither positive or negative)
 - Option1 = integrated = +5 (maximum), all others = 0
 - Option2 = autonomy = +5, all others = 0
 - Option3 = direct_manag = +5, all others = 0
 - Option4 = training = +5 (this was the one of the least scored values in the original data), all others = 0

The results are provided in Table 16.

Table 16 Predictions

```
Group 1, Option 0 (workleng=3, all = 0): -0.024
Group 2, Option 0 (workleng=23, all = 0): 0.126

Group 1, Option 1 (workleng=3, integrated = +5): 6.150
Group 2, Option 1 (workleng=23, integrated = +5): 6.300

Group 1, Option 2 (workleng=3, autonomy = +5): 5.451
Group 2, Option 2 (workleng=23, autonomy = +5): 5.600

Group 1, Option 3 (workleng=3, direct_manag = +5): 5.430
Group 2, Option 3 (workleng=23, direct_manag = +5): 5.579

Group 1, Option 4 (workleng=3, training = +5): 5.045
Group 2, Option 4 (workleng=23, training = +5): 5.195
```

The results indicate that when all other predictive variables remain neutral (neither positive or negative), those that have been at the company the least have lower overall OC score than those that have been with the company the longest. These differences are seen across all the options. In addition, *integrated* (whether the employee felt to be part of a team), had the most influence on overall OC score compared to *autonomy* (feeling trusted to do their job independently), *direct_mang* (having a positive relationship with their direct manager) and *training* (feel they get enough training to do their job effectively).

5 Discussion, Conclusion and Recommendation

5.1 Discussion

The results of the data science project highlight key areas for improvement within the organisational climate. The aim was to identify dimensions of the Organisational Climate Measure (OCM) where employees reported dissatisfaction, examine the relationship between organisational climate and factors such as work location, tenure, and management responsibilities and predict which factors most influence the overall OC score. The dimensions *integrated* (feeling part of a team) and *training* scored the lowest, suggesting that employees may feel disconnected from their teams and lack adequate training to perform effectively. These findings indicate potential areas of disconnection within the organisation, particularly around team integration and professional development support.

Analysis of overall OC score in relation to work location, management responsibilities, intention to quit, and employee sentiment revealed no statistically significant differences between groups. This means that we cannot conclude with certainty that these factors impact the organisational climate. However, the low power of these tests implies that the sample size may have been insufficient to detect small effects, so a larger sample size could provide more reliable insights. Notably, a significant difference in overall OC score was observed between employees who have been with the organisation for 13 years compared to those with 8 years, with longer-tenured employees reporting higher OC scores. This finding suggests that tenure may influence perceptions of the organisational climate, with longer-tenured employees possibly feeling more engaged or integrated.

The multiple linear regression analysis further revealed that several predictors, including tenure, involvement in decision-making, autonomy, relationship with direct managers, feeling part of a team, welfare, training, work-life balance, and organisational response to customers, significantly contribute to predicting overall OC scores. Among these, feeling part of a team had the most substantial influence on overall OC, followed by autonomy and having a positive relationship with one's direct manager. This suggests that employees' sense of belonging and trust in their roles are strong determinants of their perception of organisational climate.

Interestingly, the data showed no significant relationship between intent to quit and overall OC score. Those intending to quit did not necessarily report the most negative perceptions, suggesting that factors beyond organisational climate, such as career development or personal circumstances, may influence their decision. Additionally, a counterintuitive result was found, where employees with positive sentiment scores had lower overall OC scores, which may reflect discrepancies between how sentiment was coded from textual responses and the OC scores derived from the validated OCM.

5.2 Recommendations

Based on these findings, it is recommended that ServiceFirst focus efforts on fostering a stronger sense of team integration and providing comprehensive training opportunities. Since *integrated* was identified as both a low-scoring dimension and a significant predictor of overall OC, the organisation should prioritise initiatives that enhance team cohesion. This can include promoting open communication, encouraging cross-functional collaboration, holding regular team-building activities and fostering an inclusive culture that values diversity and

respects individual contributions. For new employees, implementing a strong onboarding process is essential to help them quickly feel part of the team and understand their role in the organisation's success.

Given that *training* also scored low, ServiceFirst should ensure that regular training and development opportunities are provided, particularly for newly recruited employees. This could involve offering job-specific training programs, ongoing skill development workshops and access to resources that help employees feel competent and confident in their roles. Continuous learning opportunities not only equip employees with necessary skills but also contribute to a positive perception of organisational support, which can enhance overall job satisfaction and retention.

To improve future surveys and analyses, ServiceFirst should consider revising the Likert scale used in the OCM assessment. A scale from 1 (very dissatisfied) to 5 (very satisfied) may yield more interpretable averages than the current -5 to +5 range, which might obscure subtle differences in satisfaction levels. Additionally, increasing the sample size would provide greater statistical power, helping to clarify relationships between OC scores and other variables and narrowing confidence intervals for more precise estimates.

5.3 Conclusion

In conclusion, the data science analysis provides valuable insights into areas of improvement within ServiceFirst's organisational climate, particularly in relation to team integration and training. While factors such as work location and management responsibilities did not show significant effects on overall OC scores, tenure was identified as a potentially influential factor, with longer-tenured employees reporting higher satisfaction. The findings underscore the importance of fostering a sense of belonging, providing robust training, and supporting employees in their roles. Implementing these recommendations can help ServiceFirst create a more positive work environment, ultimately enhancing employee retention and reducing recruitment costs. By refining survey methods and expanding sample sizes in future analyses, the organisation can gain even clearer insights into employee needs and continue building a supportive and engaged workforce.

6 References

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